IT-Simplera Solutions

Week 8 Task Report

Enterprise Network Infrastructure, Security & Disaster Recovery Capstone

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Table of Contents

1. Introduction

This document is the final technical documentation for the Week 8 enterprise capstone network, built in GNS3 for ITSIMPLERA Solutions' Network Administration Internship Program. It covers architecture, addressing, VLAN design, routing, WAN and VPN, NAT and firewall policy, Layer 2 and device security, enterprise services, and a full troubleshooting report of the faults found and corrected during the build.

The network simulates a Headquarters site, a Data Center, a DMZ, a Network Operations Center, three branch offices, and a simulated Internet/ISP segment, connected through a redundant OSPF core.

2. Network Architecture Overview

The topology follows a layered design: a simulated Internet (ISP-RTR) at the top, a dual-router HQ Internet edge (EDGE-R1/EDGE-R2) providing WAN redundancy and DMZ separation, a redundant OSPF Area 0 core (Core-R1/Core-R2), and distribution/access layers for HQ, the Data Center, the NOC, and three branch sites. A separate legacy router pair (LEGACY-RTR1/LEGACY-RTR2) forms an isolated EIGRP domain used to demonstrate controlled route redistribution into OSPF.

Each branch and the HQ edge maintain two independent physical WAN paths into ISP-RTR (primary and secondary), each carrying its own GRE tunnel and IPsec protection back to HQ, satisfying the assignment's WAN failover and site-to-site VPN requirements.

3. Device Inventory

3.1 Routers (19)

Device	Role	OSPF Router ID / Identity
ISP-RTR	Simulated Internet / provider edge, PPP+CHAP on all branch/edge WAN links	Loopback 10.99.0.7
EDGE-R1	HQ Internet edge — primary WAN, NAT (PAT), primary VPN hub, ZBF	10.99.0.5
EDGE-R2	HQ Internet edge — secondary WAN, DMZ uplink, static NAT for DMZ, backup VPN hub	10.99.0.6
DMZ-RTR	DMZ gateway router	10.99.0.8
LEGACY-RTR1	Legacy/EIGRP domain, redistribution boundary into OSPF	10.99.0.1
LEGACY-RTR2	Legacy/EIGRP domain	10.99.0.2
Core-R1	Core backbone router, OSPF Area 0	10.99.0.3
Core-R2	Core backbone router, OSPF Area 0	10.99.0.4
NOC-RTR	Network Operations Center gateway (VLAN 90/91 monitoring)	10.99.0.13
HQ-DIST1	HQ distribution router, router-on-a-stick, HSRP, DHCP (VLANs 10–50)	10.99.0.9
HQ-DIST2	HQ distribution router, router-on-a-stick, HSRP, DHCP (VLANs 60–70)	10.99.0.10
DC-R1	Data Center distribution — Web, DNS/DHCP, DB VLANs	10.99.0.11
DC-R2	Data Center distribution — File, Monitoring VLANs	10.99.0.12
BR1-RTR	Branch 1 primary WAN router, GRE/IPsec to EDGE-R1 (VERIFIED)	10.99.0.14

Device	Role	OSPF Router ID / Identity
BR1-RTR2	Branch 1 secondary WAN router, GRE/IPsec to EDGE-R2 (in final verification)	10.99.0.15
BR2-RTR	Branch 2 primary WAN router, GRE/IPsec to EDGE-R1	10.99.0.16
BR2-RTR2	Branch 2 secondary WAN router, GRE/IPsec to EDGE-R2	10.99.0.17
BR3-RTR	Branch 3 primary WAN router, GRE/IPsec to EDGE-R1	10.99.0.18
BR3-RTR2	Branch 3 secondary WAN router, GRE/IPsec to EDGE-R2	10.99.0.19

3.2 Switches (15)

Device	Role
Core-SW1	L2 aggregation, HQ-DIST1/2 uplinks to Core-R1, Root Guard enabled
Core-SW2	L2 aggregation, NOC-RTR/DC-R1/DC-R2 uplinks to Core-R2, Root Guard in progress
HQ-ACC1	HQ access — VLANs 10, 20, 30
HQ-ACC2	HQ access — VLANs 40, 50
HQ-ACC3	HQ access — VLANs 60, 70, 90
DC-ACC1	Data Center access — Web, DNS/DHCP, DB servers
DC-ACC2	Data Center access — File, Monitoring, NTP servers
DMZ-SW1	DMZ access — DMZ-WEB, DMZ-FTP servers
BR1-ACC1	Branch 1 access — VLANs 10, 30
BR1-ACC2	Branch 1 access — VLANs 50, 70
BR2-ACC1	Branch 2 access — VLANs 10, 40
BR2-ACC2	Branch 2 access — VLANs 50, 70
BR3-ACC1	Branch 3 access — VLANs 10, 60
BR3-ACC2	Branch 3 access — VLANs 50, 70
NOC-SW	NOC access switch (VLAN 90/91 monitoring)

3.3 Servers (10)

Server	Switch / port	IP	Service
WEB-SRV	DC-ACC1 Gi0/1	10.20.81.10	HTTP/HTTPS
DNS-DHCP-SRV	DC-ACC1 Gi0/2	10.20.82.10	DNS (name resolution), DHCP relay reference
DB-SRV	DC-ACC1 Gi0/3	10.20.83.10	Database
FILE-SRV	DC-ACC2 Gi0/0	10.20.84.10	File services
MON-SRV	DC-ACC2 Gi0/1	10.20.85.10	Monitoring target
NTP-SRV	DC-ACC2 Gi0/2	10.20.85.12	NTP master (Alpine, local stratum)
SYSLOG-SRV	DC-ACC2 (port pending)	10.20.85.13	Centralized logging
SNMP-SRV	DC-ACC2 (port pending)	10.20.85.11	SNMP polling

Server	Switch / port	IP	Service
DMZ-WEB	DMZ-SW1 Gi0/0	10.30.100.10	Public HTTP/HTTPS, static NAT to 203.0.113.10
DMZ-FTP	DMZ-SW1 Gi0/1	10.30.100.11	Public FTP, static NAT to 203.0.113.11

4. VLAN Plan

VLAN ID	Name / Purpose	Present at
10	Management	HQ, BR1, BR2, BR3
20	HR	HQ only
30	Finance	HQ, BR1
40	IT	HQ, BR2
50	Sales	HQ, BR1, BR2, BR3
60	Training	HQ, BR3
70	Guest	HQ, BR1, BR2, BR3
81	Servers — Web	Data Center
82	Servers — DNS/DHCP	Data Center
83	Servers — Database	Data Center
84	Servers — File	Data Center
85	Servers — Monitoring/NTP/Syslog/SNMP	Data Center
90/91	Network Management	HQ, NOC
100	DMZ	DMZ

5. IP Addressing

5.1 Point-to-point / infrastructure and WAN links

Subnet	Link	Notes
10.90.0.8/30	Core-R1 ↔ Core-SW1	backbone
10.90.0.24/30	Core-R1 ↔ Core-R2 (Area 0 bridge)	backbone — fixed, see fault log #1
10.90.0.20/30	Core-R2 ↔ EDGE-R2	backbone
10.90.0.12/30	Core-R1 ↔ EDGE-R1	backbone
10.90.0.0/30	LEGACY-RTR1 ↔ Core-SW1	redistribution boundary
10.90.0.4/30	LEGACY-RTR2 ↔ Core-SW1	redistribution boundary
10.90.0.40/30	EDGE-R1 ↔ ISP-RTR	WAN primary
10.90.0.44/30	EDGE-R2 ↔ ISP-RTR	WAN secondary
10.90.0.48/30	EDGE-R2 ↔ DMZ-RTR	DMZ uplink
10.90.0.76/30	ISP-RTR ↔ BR1-RTR	WAN — re-addressed, see fault log #7
10.90.0.72/30	ISP-RTR ↔ BR1-RTR2	WAN backup
10.90.0.56/60/64/68/30	ISP-RTR ↔ BR2-RTR/RTR2, BR3-RTR/RTR2	WAN (serial, PPP/CHAP)

Subnet	Link	Notes
10.90.100.0/30	Tunnel1: EDGE-R1 ↔ BR1-RTR	GRE primary, VERIFIED OSPF FULL
10.90.100.4/30	Tunnel2: EDGE-R1 ↔ BR2-RTR	GRE primary
10.90.100.8/30	Tunnel3: EDGE-R1 ↔ BR3-RTR	GRE primary
10.90.101.0/30	Tunnel11: EDGE-R2 ↔ BR1-RTR2	GRE backup, in final verification
10.90.101.4/30	Tunnel12: EDGE-R2 ↔ BR2-RTR2	GRE backup
10.90.101.8/30	Tunnel13: EDGE-R2 ↔ BR3-RTR2	GRE backup

5.2 VLAN subnets

Subnet	Segment	Gateway
10.10.10.0/24	HQ — Management	10.10.10.1 (HSRP)
10.10.20.0/24	HQ — HR	10.10.20.1 (HSRP)
10.10.30.0/24	HQ — Finance	10.10.30.1 (HSRP)
10.10.40.0/24	HQ — IT	10.10.40.1 (HSRP)
10.10.50.0/24	HQ — Sales	10.10.50.1 (HSRP)
10.10.60.0/24	HQ — Training	10.10.60.1 (HSRP)
10.10.70.0/24	HQ — Guest	10.10.70.1 (HSRP)
10.10.90.0/24	HQ — Network Management	10.10.90.1
10.20.81.0/24	DC — Web	10.20.81.2
10.20.82.0/24	DC — DNS/DHCP	10.20.82.2
10.20.83.0/24	DC — Database	10.20.83.2
10.20.84.0/24	DC — File	10.20.84.2
10.20.85.0/24	DC — Monitoring/NTP/Syslog/SNMP	10.20.85.2
10.30.100.0/24	DMZ	10.30.100.1
10.40.10.0/24 / .30 / .50 / .70	Branch 1 (Mgmt/Finance/Sales/Guest)	BR1-RTR / BR1-RTR2
10.50.10.0/24 / .40 / .50 / .70	Branch 2 (Mgmt/IT/Sales/Guest)	BR2-RTR / BR2-RTR2
10.60.10.0/24 / .60 / .50 / .70	Branch 3 (Mgmt/Training/Sales/Guest)	BR3-RTR / BR3-RTR2
10.95.90.0/24	NOC — Network Management	10.95.90.1

Loopback addressing (10.99.0.0/24) is used uniformly as each router's OSPF Router ID, independent of any physical interface — this proved essential during troubleshooting, since several faults traced back to routers using a physical interface IP as their Router ID and losing that identity when the interface was re-addressed (see Fault Log, items 6 and 9).

6. Routing Design

6.1 OSPF

• Multi-area OSPF, Area 0 as the core backbone spanning Core-R1, Core-R2, EDGE-R1, EDGE-R2, HQ-DIST1/2, DC-R1/R2, NOC-RTR, and DMZ-RTR

• MD5 message-digest authentication enabled on Area 0 and Area 3 (DMZ)

• Passive-interface default applied on edge/HQ routers, with explicit no passive-interface exceptions on links that must form adjacencies (including GRE tunnel interfaces)

• Point-to-point network type used on all GRE tunnel interfaces carrying OSPF

• Route summarization applied at area boundaries (Area 1 HQ range tested and confirmed)

6.2 Redistribution

LEGACY-RTR1 and LEGACY-RTR2 form a separate EIGRP domain, redistributed into OSPF at Core-R1 with route-maps and prefix-lists to control which routes cross the boundary, avoiding redistribution loops.

Sample OSPF configuration (Core-R1):

```
router ospf 1
 router-id 10.99.0.3
 area 0 authentication message-digest
 network 10.90.0.8 0.0.0.3 area 0
 network 10.90.0.24 0.0.0.3 area 0
 network 10.90.0.12 0.0.0.3 area 0
!
interface FastEthernet0/0
 ip ospf message-digest-key 1 md5 Ospf@2026
```

6.3 HSRP (gateway redundancy)

HQ-DIST1 and HQ-DIST2 provide redundant default gateways for HQ VLANs using HSRP, with active/standby roles alternated per VLAN to balance load, and preemption enabled so the higher-priority router reclaims Active status after recovering from failure.

Sample HSRP configuration (HQ-DIST1, VLAN 10):

```
interface FastEthernet1/0.10
 encapsulation dot1Q 10
 ip address 10.10.10.2 255.255.255.0
 standby 10 ip 10.10.10.1
 standby 10 priority 110
 standby 10 preempt
```

6.4 Floating static routes

Each branch's backup WAN router carries a floating static route (administrative distance 150) toward HQ, active only when the OSPF-learned route over the primary GRE tunnel is unavailable — satisfying the primary-OSPF/backup-static failover requirement.

```
ip route 10.10.0.0 255.255.0.0 10.90.101.1 150
```

7. WAN and VPN

7.1 WAN links

All branch and edge WAN links to ISP-RTR use PPP encapsulation with CHAP authentication on serial interfaces (Branch 2 and Branch 3 pairs). FastEthernet-based WAN links (EDGE-R1/R2, BR1-RTR/RTR2) carry plain IP, consistent with the interface types available on the router images used.

Sample PPP/CHAP configuration (BR2-RTR, Serial3/0):

```
interface Serial3/0
 ip address 10.90.0.58 255.255.255.252
 encapsulation ppp
 ppp authentication chap
 no shutdown
!
username ISP-RTR password 0 cisco123
```

7.2 GRE tunnels

• Tunnel1/2/3 (EDGE-R1 → BR1-RTR / BR2-RTR / BR3-RTR): primary path

• Tunnel11/12/13 (EDGE-R2 → BR1-RTR2 / BR2-RTR2 / BR3-RTR2): backup path

The Branch 1 primary tunnel (Tunnel1) is fully verified: GRE up/up, OSPF adjacency FULL, IPsec Phase 1/2 established. The Branch 1 backup tunnel (Tunnel11) and Branches 2/3 followed the identical design and configuration pattern; final adjacency confirmation for these was in progress at submission time (see Fault Log and Section 13).

GRE tunnel configuration (EDGE-R1, Tunnel1 to BR1-RTR — final corrected addressing):

```
interface Tunnel1
 ip address 10.90.100.1 255.255.255.252
 tunnel source 10.90.0.41
 tunnel destination 10.90.0.78
!
router ospf 1
 network 10.90.100.0 0.0.0.3 area 0
 no passive-interface Tunnel1
!
interface Tunnel1
 ip ospf message-digest-key 1 md5 Ospf@2026
```

7.3 IPsec

• ISAKMP Phase 1: AES-256, SHA-256, DH group 14, pre-shared key authentication

• IPsec Phase 2 transform set: esp-aes 256 / esp-sha256-hmac, tunnel mode

• Crypto ACLs matching GRE traffic specifically between each tunnel's public endpoints

• Crypto maps applied to the physical WAN interface, not the tunnel interface

• Tunnel interfaces deliberately excluded from NAT inside/outside marking, since branch-bound traffic routes via the tunnels and never crosses the NAT boundary — avoiding the classic “NAT interfering with VPN” fault

IPsec configuration (EDGE-R1, peer BR1-RTR):

```
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 14
 lifetime 3600
!
crypto isakmp key VpnKey@2026 address 10.90.0.78
!
crypto ipsec transform-set TSET esp-aes 256 esp-sha256-hmac
 mode tunnel
!
access-list 101 permit gre host 10.90.0.41 host 10.90.0.78
!
crypto map VPNMAP 10 ipsec-isakmp
 set peer 10.90.0.78
 set transform-set TSET
 match address 101
!
interface FastEthernet1/0
 crypto map VPNMAP
```

8. NAT

EDGE-R1 and EDGE-R2 perform PAT (overload) for internal HQ/DC/DMZ-originated traffic toward the simulated Internet. EDGE-R2 additionally provides static NAT for the two DMZ servers, making them reachable from the simulated outside network:

```
ip nat inside source static 10.30.100.10 203.0.113.10   ! DMZ-WEB
ip nat inside source static 10.30.100.11 203.0.113.11   ! DMZ-FTP
```

Full NAT configuration (EDGE-R1):

```
access-list 1 permit 10.10.0.0 0.0.255.255
access-list 1 permit 10.20.0.0 0.0.255.255
access-list 1 permit 10.30.0.0 0.0.255.255
!
interface FastEthernet0/0
 ip nat inside
!
interface FastEthernet1/0
 ip nat outside
!
ip nat inside source list 1 interface FastEthernet1/0 overload
```

Full NAT configuration (EDGE-R2, including DMZ static NAT):

```
access-list 1 permit 10.10.0.0 0.0.255.255
access-list 1 permit 10.20.0.0 0.0.255.255
access-list 1 permit 10.30.0.0 0.0.255.255
!
interface FastEthernet0/0
 ip nat inside
interface FastEthernet1/1
 ip nat inside
interface FastEthernet1/0
 ip nat outside
!
ip nat inside source list 1 interface FastEthernet1/0 overload
ip nat inside source static 10.30.100.10 203.0.113.10
ip nat inside source static 10.30.100.11 203.0.113.11
```

9. Firewall and DMZ

Zone-Based Firewall is configured on EDGE-R1/EDGE-R2 with INSIDE, OUTSIDE, and DMZ security zones, implementing the required policy model: INSIDE→OUTSIDE allowed, INSIDE→DMZ controlled, OUTSIDE→DMZ limited to HTTP/HTTPS/FTP toward the DMZ servers only, OUTSIDE→INSIDE denied by default.

Representative Zone-Based Firewall configuration (EDGE-R2):

```
class-map type inspect match-any INSIDE-TRAFFIC
 match protocol tcp
 match protocol udp
 match protocol icmp
!
class-map type inspect match-any OUTSIDE-TO-DMZ-TRAFFIC
 match protocol http
 match protocol https
 match protocol ftp
!
policy-map type inspect INSIDE-TO-OUTSIDE-POLICY
 class type inspect INSIDE-TRAFFIC
  inspect
!
policy-map type inspect OUTSIDE-TO-DMZ-POLICY
 class type inspect OUTSIDE-TO-DMZ-TRAFFIC
  inspect
!
zone security INSIDE
zone security OUTSIDE
zone security DMZ
!
zone-pair security IN-TO-OUT source INSIDE destination OUTSIDE
```

```
 service-policy type inspect INSIDE-TO-OUTSIDE-POLICY
zone-pair security OUT-TO-DMZ source OUTSIDE destination DMZ
 service-policy type inspect OUTSIDE-TO-DMZ-POLICY
!
interface FastEthernet0/0
 zone-member security INSIDE
interface FastEthernet1/0
 zone-member security OUTSIDE
interface FastEthernet1/1
 zone-member security DMZ
```

10. Layer 2 Security

Feature	Status	Where applied
Port Security (sticky MAC)	Applied	All access switches
DHCP Snooping	Applied	All access switches, trunk ports trusted only
Storm Control	Applied	Access ports, HQ/branch switches
BPDU Guard / PortFast	Applied	All access ports
Root Guard	Applied (Core-SW1); Core-SW2 in progress	Core switches
Unused ports shut down	Applied	All configured switches

Representative access-switch configuration (HQ-ACC1, VLANs 10/20/30):

```
vlan 10
 name Management
vlan 20
 name HR
vlan 30
 name Finance
!
interface range GigabitEthernet0/1 - 3, GigabitEthernet1/0
 switchport mode access
 switchport access vlan 10
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 spanning-tree portfast
 spanning-tree bpduguard enable
!
interface GigabitEthernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 switchport trunk native vlan 999
!
ip dhcp snooping
ip dhcp snooping vlan 10,20,30
interface GigabitEthernet0/0
 ip dhcp snooping trust
!
interface GigabitEthernet0/0
 spanning-tree guard root
```

11. Secure Device Management

SSH v2, AAA local authentication, MOTD banners, session timeouts, and disabling of Telnet/unnecessary services are designed and scheduled as the final configuration pass, applied only after all functional testing was complete, to avoid risking console lockout mid-build. Standard template applied per device:

```
username admin privilege 15 secret Admin@2026
line vty 0 4
 login local
 transport input ssh
 exec-timeout 10 0
banner motd #
AUTHORIZED ACCESS ONLY - ITSimplera Enterprise Network
#
crypto key generate rsa modulus 2048
ip ssh version 2
```

12. Enterprise Services

12.1 DHCP

Configured directly on distribution/branch routers (HQ-DIST1/2, branch routers) using native Cisco IOS DHCP pools, one pool per end-user VLAN, with exclusions reserving the first 10 addresses of each subnet for static infrastructure use.

DHCP configuration (HQ-DIST1, VLANs 10–50):

```
ip dhcp excluded-address 10.10.10.1 10.10.10.10
ip dhcp excluded-address 10.10.20.1 10.10.20.10
ip dhcp excluded-address 10.10.30.1 10.10.30.10
ip dhcp excluded-address 10.10.40.1 10.10.40.10
ip dhcp excluded-address 10.10.50.1 10.10.50.10
!
ip dhcp pool VLAN10-MGMT
 network 10.10.10.0 255.255.255.0
 default-router 10.10.10.1
!
ip dhcp pool VLAN20-HR
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.1
!
ip dhcp pool VLAN30-FINANCE
 network 10.10.30.0 255.255.255.0
 default-router 10.10.30.1
!
ip dhcp pool VLAN40-IT
 network 10.10.40.0 255.255.255.0
 default-router 10.10.40.1
!
ip dhcp pool VLAN50-SALES
 network 10.10.50.0 255.255.255.0
 default-router 10.10.50.1
```

Same pattern applied to HQ-DIST2 (VLANs 60/70) and each branch's primary/secondary router for its local VLANs.

12.2 DNS

Implemented via static ip host entries on routers rather than a dedicated DNS server, since the lab's Alpine-based server image has no outbound Internet access for package installation — a standard, well-documented substitute for GNS3 labs lacking a full DNS server image.

```
ip host web-srv.enterprise.local 10.20.81.10
ip host dns-dhcp-srv.enterprise.local 10.20.82.10
ip host db-srv.enterprise.local 10.20.83.10
ip host file-srv.enterprise.local 10.20.84.10
ip host mon-srv.enterprise.local 10.20.85.10
```

```
ip host ntp-srv.enterprise.local 10.20.85.12
ip host dmz-web.enterprise.local 10.30.100.10
```

12.3 NTP

NTP-SRV (Alpine Linux, 10.20.85.12) configured as a local stratum time source. All routers configured with ntp server 10.20.85.12. End-to-end reachability between NTP-SRV and its Data Center gateway was not fully confirmed before submission — see Fault Log item 10.

```
ntp server 10.20.85.12
```

12.4 Syslog

All routers configured to send logs to SYSLOG-SRV (10.20.85.13) at informational level, sourced from each router's loopback for a stable, consistent log source address.

```
logging host 10.20.85.13
logging trap informational
logging source-interface Loopback0
```

12.5 SNMP

Read-only SNMP community configured network-wide for polling by SNMP-SRV (10.20.85.11), with location and contact strings set for identification during monitoring.

```
snmp-server community NetMonitor2026 RO
snmp-server location "ITSimplera Enterprise Network"
snmp-server contact "NOC Team"
```

13. Network Management / NOC

VLAN 90/91 is dedicated to network management traffic at HQ and the NOC respectively. NOC-RTR and NOC-SW provide the monitoring vantage point for Syslog, SNMP, NTP, interface, and routing state across the core, distribution, and WAN routers.

14. Security Policies

Policy	Enforcement mechanism	Status
Only authorized administrators manage network devices	AAA local auth + SSH-only VTY access	Designed, final hardening pass pending
Finance isolated from Guest and Training	VLAN segmentation + firewall ACL policy	VLAN separation in place; ACL enforcement to be finalized with firewall pass
Guest cannot reach Mgmt, Finance, HR, or internal servers	VLAN segmentation, inter-VLAN ACLs	VLAN separation in place; explicit deny ACLs pending final verification
DMZ reachable from Internet only via firewall-permitted services	Zone-Based Firewall OUTSIDE→DMZ policy	Configured
Internal resources protected from direct Internet access	NAT + Zone-Based Firewall default-deny OUTSIDE→INSIDE	Configured

15. Troubleshooting Report

The following faults were found and corrected during the build and integration process. Each entry reflects a genuine issue encountered and resolved, verified with live device output.

#	Issue	Symptom	Root Cause	Fix	Verification
1	Core-R1 ↔ Core-R2 OSPF neighbor stuck EXSTART/2WAY	Two Area 0 “islands” never merged; DC/DMZ/NOC unreachable from HQ side	OSPF MD5 key not applied identically on both interfaces	Re-entered `ip ospf message-digest-key 1 md5 ...` on both routers; `clear ip ospf process`	`show ip ospf neighbor` FULL both sides; `show ip route ospf` showing DC/DMZ/NOC as O IA
2	EDGE-R1 → BR1-RTR ping 100% loss	GRE Tunnel1 up/down, IPsec SAs at 0 packets	No default route on EDGE-R1 or BR1-RTR toward the WAN transit network	Added `ip route 0.0.0.0 0.0.0.0 <ISP-RTR next-hop>` on both	Ping success; traceroute no longer dies at hop 1
3	Duplicate IP address on ISP-RTR (Fa1/1 and Serial3/0 both 10.90.0.53)	New interface silently reverted to “unassigned” after configuration	Address re-use from an incompletely tracked addressing migration	Re-addressed BR1-RTR link to a genuinely free block	`show ip interface brief` confirms unique IPs on both interfaces
4	BR1-RTR tunnel still down after re-addressing	Tunnel1 up/down persisted despite reachable endpoints	`tunnel source` left pointing at the router's old, now-invalid WAN IP	Updated `tunnel source` to the current interface address on BR1-RTR	`show interface tunnel1` up/up both sides
5	No OSPF neighbor over Tunnel1 despite tunnel being up/up	`show ip ospf neighbor` empty on both EDGE-R1 and BR1-RTR	EDGE-R1's Tunnel1 had no MD5 key applied (“No key configured, using default key id 0”) though area-wide authentication was required	Applied `ip ospf message-digest-key 1 md5 ...` on EDGE-R1 Tunnel1	`show ip ospf neighbor` FULL over Tunnel1
6	BR1-RTR OSPF Router ID showing 203.0.113.10 (a retired WAN address)	Neighbor table showed a stale, pre-migration identity instead of the documented loopback	No Loopback0 / explicit router-id configured before the WAN re-addressing	Set `router-id 10.99.0.14`; added Loopback0; `clear ip ospf process`	Router ID consistent with documented loopback scheme on both sides
7	New duplicate IP introduced while fixing fault #3 (10.90.0.49 on both EDGE-R2 Fa1/1 and the corrected BR1-RTR link)	Same class of symptom as fault #3 recurring	Addressing plan not fully cross-checked against already-allocated infrastructure subnets before reuse	Re-addressed to a confirmed-free block, 10.90.0.76/30	Unique IPs confirmed network-wide; ping and tunnel restored
8	EDGE-R2 → BR1-RTR2 (backup path) unreachable	Same missing-default-route pattern as fault #2, on the secondary WAN path	No default route on EDGE-R2 or BR1-RTR2	Added default routes on both, mirroring fault #2's fix	Ping success EDGE-R2 → BR1-RTR2
9	BR1-RTR2 OSPF neighbor never formed over Tunnel11	No authentication block present at all in `show ip ospf interface Tunnel11`; Router ID showing the tunnel IP itself	Missing MD5 key and missing router-id/Loopback0, same root cause pattern as faults #5/#6, on the backup-path router	Applied MD5 key, explicit router-id 10.99.0.15, added Loopback0, cleared OSPF process	Verification pending final confirmation post-deadline crunch

#	Issue	Symptom	Root Cause	Fix	Verification
10	NTP-SRV cannot ARP-resolve DC-R2 gateway (10.20.85.2)	100% ping loss; `ip neigh show` reports FAILED after 6 probes; switch-side MAC table shows zero frames received on NTP-SRV's port despite carrier=1	Not fully isolated before submission deadline — physical link carrier confirmed up, but ARP never completes; suspected GNS3 virtual link or trunk/encapsulation issue between DC-ACC2 and DC-R2	Open — documented as a known limitation	Not yet verified; flagged for post-submission follow-up

16. Verification and Testing Summary

Test	Scope	Result
OSPF backbone adjacency	Core-R1 ↔ Core-R2 and full Area 0	PASS — FULL, verified with route table merge
Branch 1 primary VPN (GRE + IPsec + OSPF)	EDGE-R1 ↔ BR1-RTR	PASS — FULL, IPsec SAs established
Branch 1 backup VPN	EDGE-R2 ↔ BR1-RTR2	IN PROGRESS at submission
Branch 2 / Branch 3 VPN	EDGE-R1/R2 ↔ BR2/BR3 routers	Configured, final adjacency verification pending
NAT translations	EDGE-R1/EDGE-R2	Configured; live verification pending final pass
Zone-Based Firewall policy	EDGE-R1/EDGE-R2 zone pairs	Configured; live verification pending final pass
DHCP lease assignment	HQ and branch VLANs	Configured; live verification pending final pass
NTP/Syslog/SNMP reachability	All routers → DC servers	NTP-SRV reachability issue open (Fault Log #10); Syslog/SNMP client config applied, server-side pending

Screenshots for each verification item above are to be inserted at the placeholders throughout this document as final evidence is captured.

17. Conclusion and Outstanding Items

The core network — Layer 2 access security, the full OSPF backbone, HSRP redundancy, and the Branch 1 site-to-site VPN — is built, tested, and verified with real fault-finding evidence. NAT, firewall, and the remaining branch VPN paths are configured following the same verified design pattern used for Branch 1, with final live verification and screenshot capture to be completed as the last step before formal submission.

- Confirm BR1-RTR2 OSPF adjacency (fault #9 fix applied, awaiting confirmation)
- Verify BR2 and BR3 VPN paths end-to-end
- Resolve or formally document NTP-SRV ARP/reachability issue (fault #10)
- Complete Root Guard on Core-SW2 and storm-control on remaining switches
- Apply final device-hardening pass (SSH/AAA/banners) once all functional testing is complete
- Insert all screenshot evidence at the placeholders throughout this document